

PRDA-05

CUSTOMER SHOPPING BEHAVIOUR ANALYSIS

An Interactive Power BI Analysis of Shopping Patterns Across 10 Istanbul Malls (2021–2023)

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Project Overview

This project examines customer shopping behaviour across 10 malls in Istanbul, focusing on **who is buying, what they are buying, and how they are paying**.

The aim is to turn the data into **useful business insights** that can support decisions around customers, products, marketing and payment strategies : rather than simply presenting charts.

What's Covered

- Shopping patterns by gender and age
- Best-selling product categories and the customers buying them
- Payment methods and how they vary across categories
- Power BI dashboard analysis and key business insights

KEY METRICS

252M TL

Total Revenue

99K

Total Orders

99K

Unique Customers

299K

Total Quantity

2.53K TL

Avg. Order Value

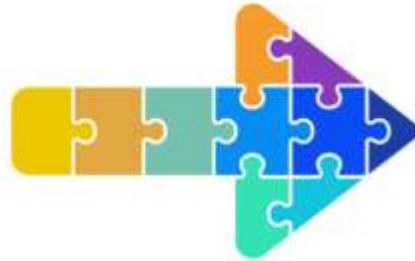
Dataset Overview

The dataset covers customer transactions from 10 shopping malls in Istanbul between **2021 and 2023**. Each row represents one purchase, including details about the customer, products purchased, amount paid, payment method and shopping location.

Field	What it means
Gender / Age	Basic customer details
Category	What was bought (8 product types)
Quantity / Price	How many items, and the price per item, in Turkish Lira
Payment method	Cash, Credit Card, or Debit Card
Mall / Date	Where and when the purchase happened

99,457

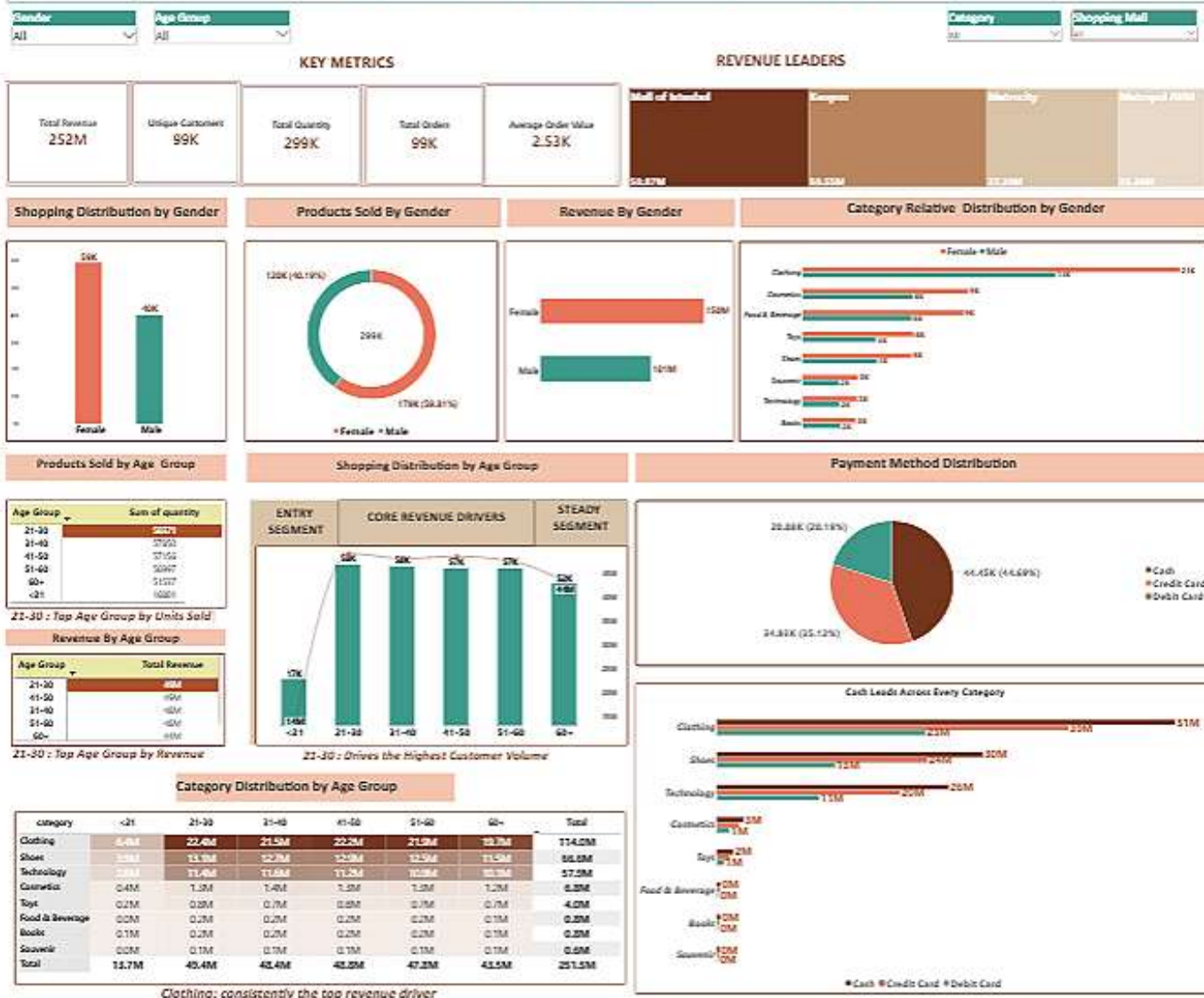
purchases in total



INTERACTIVE DASHBOARD

The interactive Power BI dashboard allows the analysis to be filtered by **gender, age group, category and shopping mall** to explore customer behaviour from different perspectives

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customer pbi1, Duplicate of Duplicate of Page 2

Public snapshot

Data updated on 4/9/26, 4:07 pm

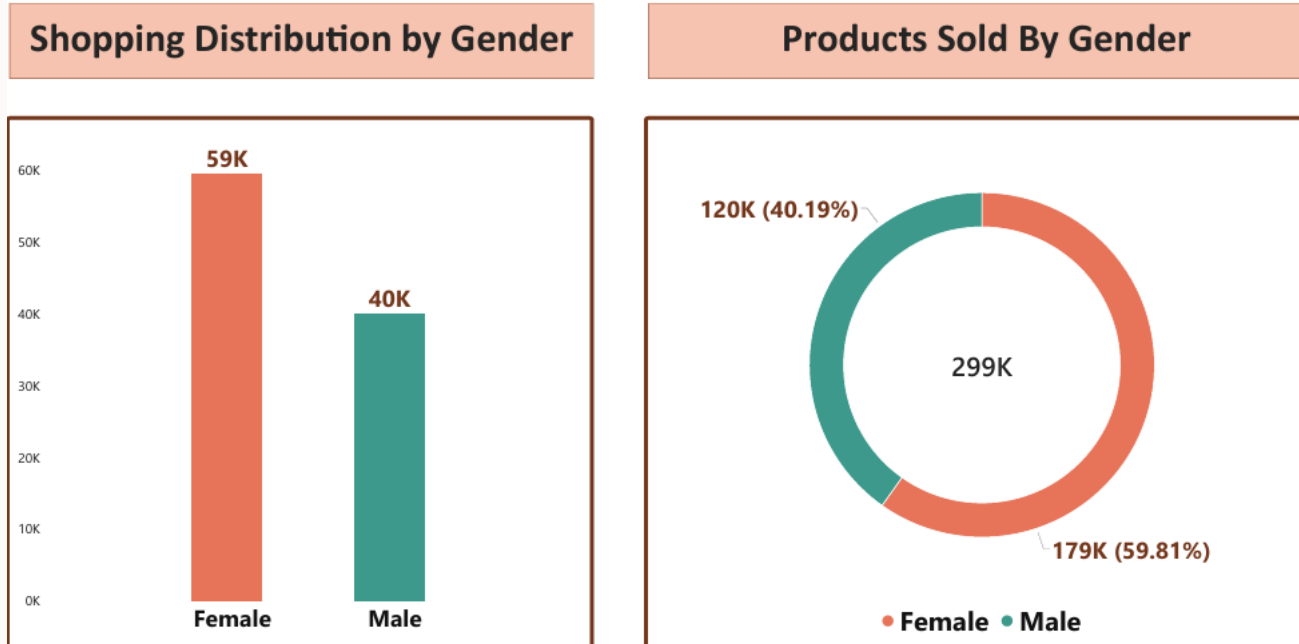


Who's Shopping More?



Q1. How is the shopping distribution according to gender?

Q2. Which gender did we sell more products to?



59.8%
of all orders placed by women

Observation Out of **99,457 total orders**, women placed **59,482 (about 60%)**, compared with **39,975 by men**. The same pattern is seen in quantity purchased, with women buying **179K units** compared with **120K units** by men

Business Insight Female customers are clearly the larger shopping segment across both orders and quantity. This makes them an important audience for customer acquisition and retention, while the male segment presents an opportunity for further growth.

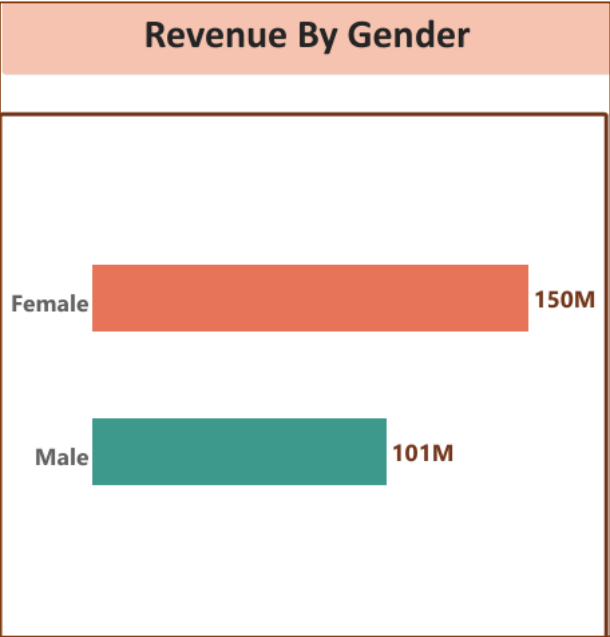
Who's Bringing in More Revenue?

Q3. Which gender generated more revenue?



59.6%

of total revenue from women



Observation

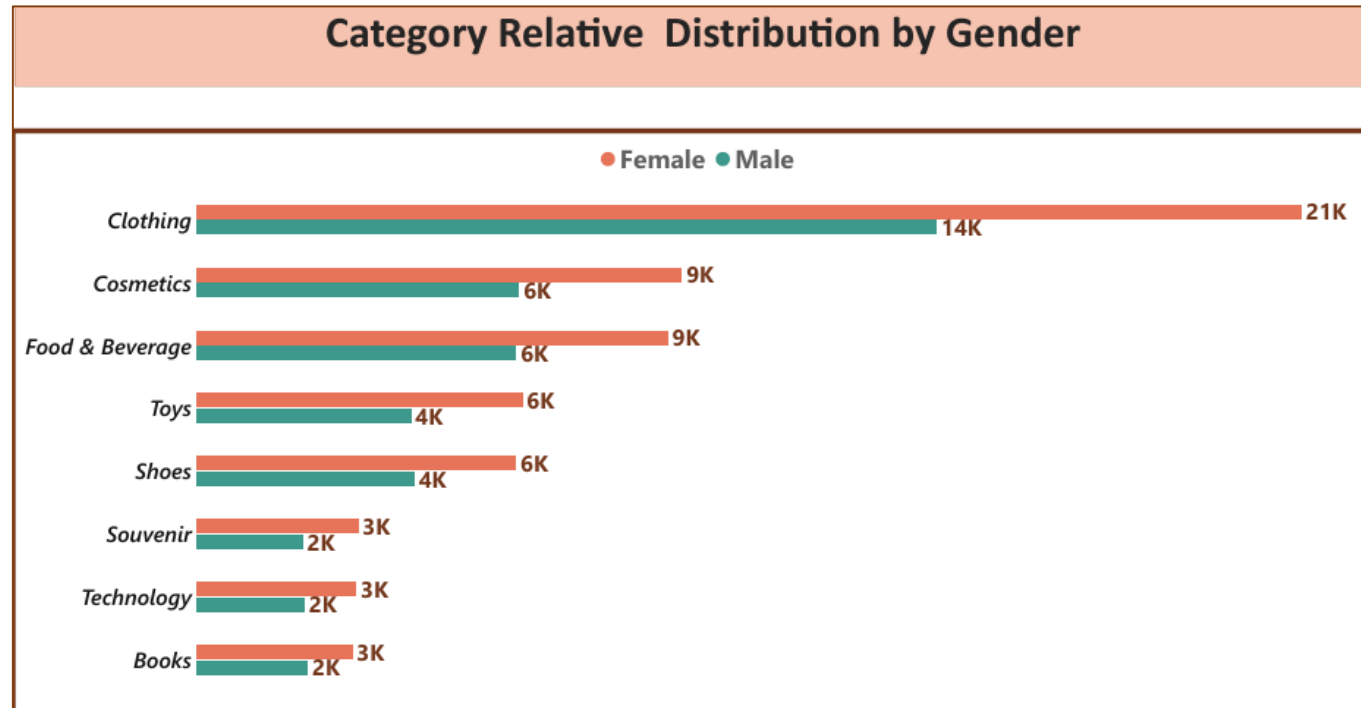
Women generated **150.2M TL** in revenue compared with **101.3M TL** for men, a difference of almost **49M TL**. This follows the same overall pattern seen in orders and quantity, with women contributing roughly **60%** of the total.

Business Insight

Female customers contribute more revenue alongside higher order volume and quantity. This suggests that **customer reach and repeat purchases** could be important areas to focus on, while the male segment provides an opportunity for further revenue growth.

Do Men and Women Buy Different Things?

Q4. Distribution of purchase categories relative to other columns (Gender).



100%

category ranking matches across genders

Observation

Clothing is the leading category for both female and male customers, with **21K units for females and 14K for males**. The remaining categories also follow a broadly similar pattern across both genders, although the volumes differ.

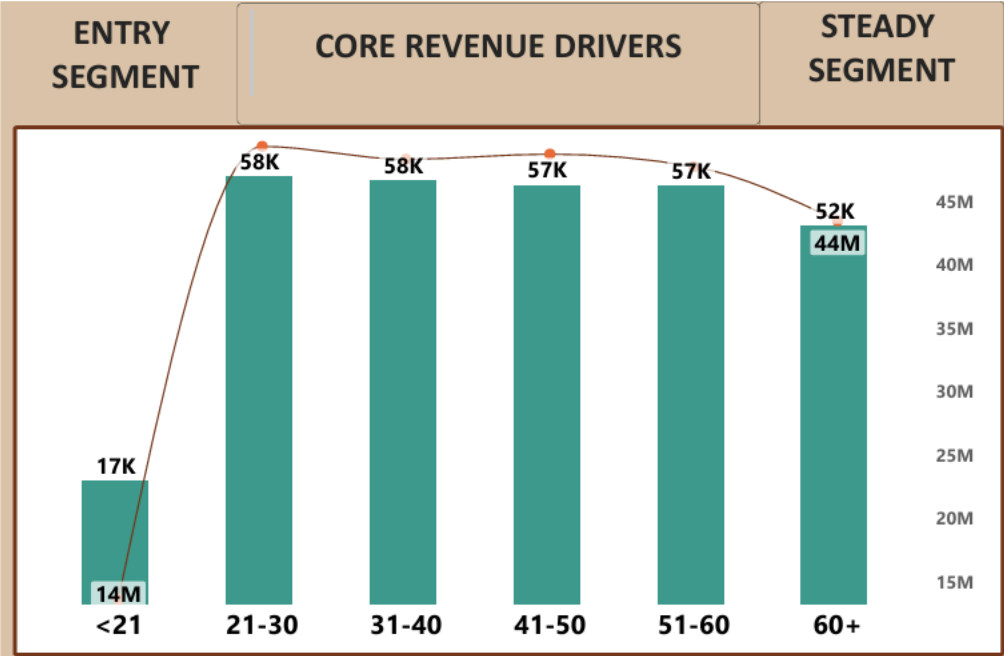
Business Insight

The similar category pattern suggests that gender does not significantly change the overall category mix. The main difference is the higher purchase volume from female customers, so the company can maintain a broadly common category strategy while using different marketing approaches to engage each gender.

Which Age Groups Shop the Most?

Q5. Shopping distribution by age. Q6. Which age category sold more products? Q7. Which age category generated more revenue?

Shopping Distribution by Age Group



Products Sold by Age Group

Age Group	Sum of quantity
21-30	58371
31-40	57850
41-50	57156
51-60	56997
60+	51537
<21	16801

Revenue By Age Group

Age Group	Total Revenue
21-30	49M
41-50	49M
31-40	48M
51-60	48M
60+	44M

+2.5%

Demand spread across the 21–60 core

Observation

21–30 comes out just ahead on both items **bought (58,371)** and **revenue (49.4M TL)**, but the difference is small. The 21–60 age groups show very similar purchase levels, while the under-21 group is clearly lower, with **16,801 items bought and 13.7M TL** in revenue.

Business Insight

the results show a fairly balanced customer base across ages 21–60 rather than one dominant age group. A broader campaign targeting this core age range could be more effective, while a separate promotion for the under-21 segment could help increase its contribution.

Does Age Change What People Buy?

Q8. Distribution of purchase categories relative to other columns (Age Group).

Category Distribution by Age Group							
category	<21	21-30	31-40	41-50	51-60	60+	Total
Clothing	6.4M	22.4M	21.5M	22.2M	21.9M	19.7M	114.0M
Shoes	3.9M	13.1M	12.7M	12.9M	12.5M	11.5M	66.6M
Technology	2.6M	11.4M	11.6M	11.2M	10.9M	10.1M	57.9M
Cosmetics	0.4M	1.3M	1.4M	1.3M	1.3M	1.2M	6.8M
Toys	0.2M	0.8M	0.7M	0.8M	0.7M	0.7M	4.0M
Food & Beverage	0.0M	0.2M	0.2M	0.2M	0.2M	0.1M	0.8M
Books	0.1M	0.2M	0.2M	0.2M	0.2M	0.1M	0.8M
Souvenir	0.0M	0.1M	0.1M	0.1M	0.1M	0.1M	0.6M
Total	13.7M	49.4M	48.4M	48.8M	47.8M	43.5M	251.5M

114 M

Clothing revenue across all ages

Observation

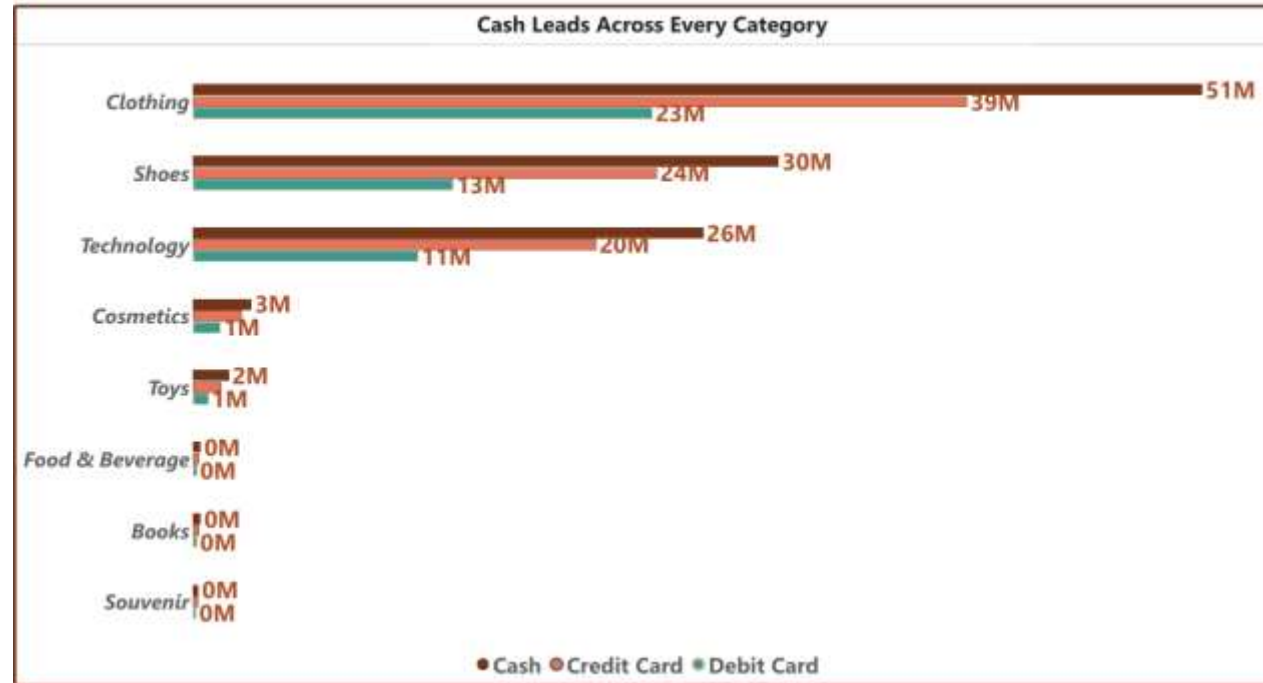
Clothing is the leading category across all age groups, followed by **Shoes and Technology**. The category ranking is broadly similar across the **21–60 age groups**, while the under-21 group records **lower revenue** across all major categories.

Business Insight

Purchase category patterns are broadly **consistent** across age groups, with **Clothing, Shoes and Technology** driving most of the revenue. This suggests the company can maintain focus on these major categories across the **21–60 customer base**, while using targeted offers to grow lower-performing categories and the **under-21 segment**

Does Age Change What People Buy?

Q9. Does the payment method have a relation with other columns?



8/8

categories where Cash leads

Observation

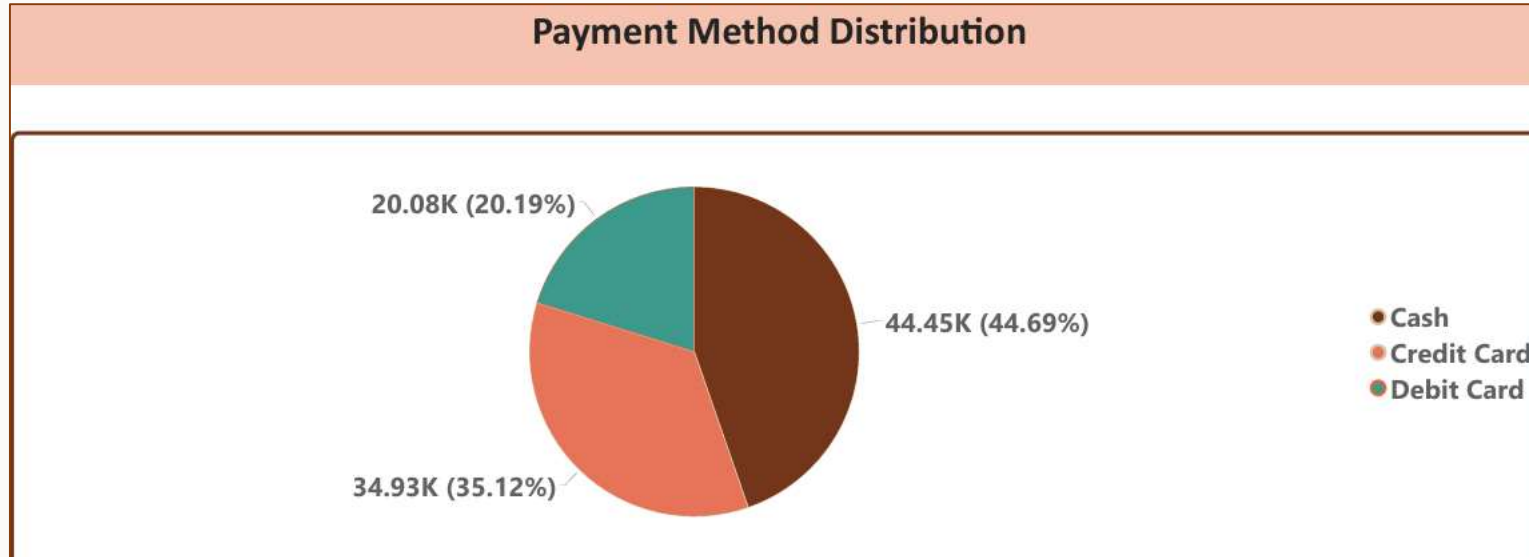
Cash comes out on top no matter what category you look at. In Clothing it's **51.3M TL** cash versus **39.4M credit** and **23.3M debit**; Shoes and Technology follow that exact same order.

Business Insight

The **Cash > Credit > Debit** ordering holds with zero exceptions across 8 independent categories - Cash leads across all product categories, followed by Credit Card and Debit Card. This consistent pattern suggests that payment preferences are broadly similar across categories, so payment initiatives can be considered across the wider customer base rather than category by category.

How Do People Pay?

Q10. How is the distribution of the payment method?



55.3%

of transactions already on cards

Observation

Cash is the most common payment method at 44.69% of transactions, followed by Credit Card at 35.12% and Debit Card at 20.19%. Together, **card payments** account for 55.31% of transactions.

Business Insight

Card payments already make up the majority of transactions, so the focus should not be on introducing digital payments but on gradually encouraging **cash users to shift towards card payments** through simple incentives or offers.

What This Data Is Really Telling Us

- *Women lead in orders, quantity and revenue, contributing around 60% of the overall business across these measures.*
- *Customers aged 21–50 form the strongest-performing age groups, while the under-21 segment is the clear outlier with much lower purchase volume and revenue.*
- *Clothing is the largest revenue contributor by a wide margin, followed by Shoes and Technology.*
- *The category mix is broadly similar across age groups and genders, with differences mainly appearing in overall purchase volumes rather than the leading categories.*
- *Cash is the most-used payment method, although card payments combined account for the majority of transactions. Cash also leads across the product categories shown.*
- *Mall of Istanbul and Kanyon are the top revenue-generating malls, performing clearly ahead of the other malls in the dashboard.*

What We'd Suggest Doing About It



**KEEP CLOTHING AS
A PRIORITY**

It contributes nearly half of total revenue, so it should remain a key focus while testing ways to grow smaller categories.

1



**TARGET THE 21–50 AGE
GROUPS TOGETHER**

These groups show similar purchase levels, making a broader campaign more practical than highly segmented campaigns. The under-21 group can be addressed separately.

2



**CONTINUE TARGETING
BOTH GENDERS**

Women contribute around 60% of orders and revenue, while men still represent about 40%. Both segments are worth targeting with different messaging.

3



**EXPLORE CLOTHING +
SHOES BUNDLES**

Together, these categories contribute around 72% of revenue, making them suitable for bundle or cross-selling opportunities.

4



**TEST DIGITAL-PAYMENT
INCENTIVES**

Cards already account for the majority of transactions. Small incentives could encourage some cash users to shift toward card payments, particularly at lower-performing malls.

5



**LEARN FROM THE TOP-
PERFORMING MALLS**

Mall of Istanbul and Kanyon lead in revenue. Understanding what drives their performance could help identify practices worth testing at other malls.

6

CONCLUSION

Looking across everything — who's buying, what they're buying, and how they're paying — the same picture keeps showing up: women aged roughly 21 to 50 are among the strongest customer groups, Clothing carries most of the revenue, and card payments already account for the majority of transactions. These findings provide a practical basis for deciding where to focus marketing, inventory and payment strategy going forward.

THANKYOU